

E851604_1

The equation $x^4 + xy + x + y^3 + 2 = 0$ has no integer solutions

Size $h = 32$, length $l = 11$

Theorem 1. *The equation*

$$x^4 + xy + x + y^3 + 2 = 0 \tag{1}$$

has no integer solutions.

The proof uses four polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), where $d_1 = H_2$, $d_2 = H_1$, $d_3 = 1$, and $d_4 = 2$, and the product formula for three Hilbert symbols. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let p be an odd prime, and let a, b, r, c, t be integers with $a \neq 0$, $b \neq 0$ and $at = r^2 - bc^2$. If $p \nmid b$, and p does not divide both r and c , then $(a, b)_p = 1$.*

Proof. If $p \nmid a$, this is Theorem 2(ii) with $\alpha = \beta = 0$. If $p \mid a$, then $r^2 \equiv bc^2 \pmod{p}$. If $p \mid c$, then $p \mid r$, which is excluded. Hence $p \nmid c$, so $b \equiv (rc')^2 \pmod{p}$, where $cc' \equiv 1 \pmod{p}$, and $(b/p) = 1$ as $p \nmid b$. Theorem 2(ii) with $\beta = 0$ gives $(a, b)_p = (b/p)^{v_p(a)} = 1$. $\square$

Lemma 4. Let a, b, r, c, t be integers with $at = r^2 - bc^2 \neq 0$ and $b \neq 0$. Then $(a, b)_v = (t, b)_v$ for every v .

Proof. The triple $(r, s, z) = (1, c, r)$ is a non-zero rational solution of $z^2 = (at)r^2 + bs^2$, so $(at, b)_v = 1$. By Theorem 2(v), $(a, b)_v(t, b)_v = (at, b)_v = 1$, and both symbols are ± 1 . $\square$

Lemma 5. Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 6. Let a, b, c be real numbers with $a > 0$, $c > 0$ and $256ac^3 > 27b^4$. Then $at^4 + bt + c > 0$ for every real number t .

Proof. For $t = 0$ this is clear. For $t \neq 0$, the inequality between the arithmetic and geometric means of at^4 , $c/3$, $c/3$ and $c/3$ gives $at^4 + c \geq 4(at^4c^3/27)^{1/4} = (256ac^3/27)^{1/4}|t| > |b||t| \geq -bt$. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + xy + x + y^3 + 2$, so that (1) is the equation $F(x, y) = 0$. For the first identity, we define

$$\begin{aligned} H_1 &= x - y, \\ T_1 &= 5x^2 + 2xy + 8x + 2y^2 + 3, \\ U_1 &= x^2 + x + 2, \\ V_1 &= x + 1, \\ G_1 &= -2, \\ \alpha_1 &= 1, \\ \beta_1 &= -x, \\ \gamma_1 &= 0, \end{aligned}$$

for the second identity, we define

$$\begin{aligned} H_2 &= x^2 + x - 3y, \\ T_2 &= 9x^4 + 28x^3 + 2x^2y + 32x^2 + 2xy + 22x + 6y^2 + 3, \\ U_2 &= 3x^3 + 2x^2 - x - 6, \\ V_2 &= 5x^2 + 8x + 3, \\ G_2 &= -18, \\ \alpha_2 &= -35x - 146, \\ \beta_2 &= 21x^2 + 68x - 70, \\ \gamma_2 &= 0, \end{aligned}$$

for the third identity, we define

$$\begin{aligned} H_3 &= -y - 1, \\ T_3 &= x + y^2 - y + 1, \\ U_3 &= x^2, \end{aligned}$$

$$\begin{aligned} V_3 &= 1, \\ G_3 &= -1, \end{aligned}$$

for the fourth identity, we define

$$\begin{aligned} H_4 &= -y - 1, \\ T_4 &= x + y^2 - y + 1, \\ U_4 &= x^2 - 1, \\ V_4 &= x, \\ G_4 &= -1, \\ \alpha_4 &= -1, \\ \beta_4 &= x, \\ \gamma_4 &= 0, \end{aligned}$$

for the auxiliary polynomials H_1 and H_2 together, we define

$$\begin{aligned} \lambda_1 &= -60x^3 - 15x^2y - 22x^2 - 15xy^2 + 32xy + 5x + 32y^2 - 3, \\ \mu_1 &= 15x^3 + 13x^2 + 9x + 1, \\ \nu_1 &= -15x + 32, \end{aligned}$$

and, for the signs, we also use

$$\begin{aligned} Q_1 &= x^2 + xy + x + y^2, \\ M_1 &= x^4 + x^3 + x^2 + x + 2, \\ E_1 &= 3x^2 + 4x, \\ Q_2 &= x^4 + 2x^3 + 3x^2y + x^2 + 3xy + 9x + 9y^2, \\ M_2 &= x^6 + 3x^5 + 30x^4 + 10x^3 + 9x^2 + 27x + 54, \\ E_2 &= 27x^4 + 54x^3 + 27x^2 + 324x, \\ L_3 &= y + 1, \\ Q_3 &= y^3 + 2, \\ \Delta_3 &= 256y^9 + 1536y^6 - 27y^4 + 2964y^3 - 162y^2 - 108y + 2021, \\ L_4 &= y + 1, \\ Q_4 &= y^3 + 2, \\ \Delta_4 &= 256y^9 + 1536y^6 - 27y^4 + 2964y^3 - 162y^2 - 108y + 2021. \end{aligned}$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 - H_2V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 - H_1V_2^2 = G_2F, \quad (3)$$

$$H_3T_3 - U_3^2 - V_3^2 = G_3F, \quad (4)$$

$$H_4T_4 - U_4^2 - 2V_4^2 = G_4F, \quad (5)$$

$$\alpha_1U_1 + \beta_1V_1 + \gamma_1F = 2, \quad (6)$$

$$\alpha_2U_2 + \beta_2V_2 + \gamma_2F = 2 \cdot 3^2 \cdot 37, \quad (7)$$

$$\alpha_4U_4 + \beta_4V_4 + \gamma_4F = 1, \quad (8)$$

$$\lambda_1H_1 + \mu_1H_2 + \nu_1F = 2^6, \quad (9)$$

$$-F = Q_1 H_1 - M_1, \quad (10)$$

$$4Q_1 = (x + 2y)^2 + E_1. \quad (11)$$

$$-27F = Q_2 H_2 - M_2, \quad (12)$$

$$36Q_2 = (3x^2 + 3x + 18y)^2 + E_2. \quad (13)$$

$$F = x^4 + L_3 x + Q_3, \quad (14)$$

$$\Delta_3 = 256Q_3^3 - 27L_3^4, \quad (15)$$

$$F = x^4 + L_4 x + Q_4, \quad (16)$$

$$\Delta_4 = 256Q_4^3 - 27L_4^4. \quad (17)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) . Put

$$\begin{aligned} A_1 &= -d_2 = -H_1, & B_1 &= -d_1 = -H_2, & T_{A_1} &= -T_1, & T_{B_1} &= -T_2, \\ A_2 &= H_3, & B_2 &= -d_3 = -1, & T_{A_2} &= T_3, \\ A_3 &= H_4, & B_3 &= -d_4 = -2, & T_{A_3} &= T_4. \end{aligned}$$

By (2)–(5), after multiplying by squares of the constants c in $d_k = cH_{k'}$ if necessary, we obtain

$$A_1 T_{A_1} = U_1^2 - B_1 V_1^2, \quad B_1 T_{B_1} = U_2^2 - A_1 V_2^2. \quad (18)$$

$$A_2 T_{A_2} = U_3^2 - B_2 V_3^2. \quad (19)$$

$$A_3 T_{A_3} = U_4^2 - B_3 V_4^2. \quad (20)$$

Step 1: signs. *The sign of H_1 .* Since $F(x, y) = 0$, (10) gives $Q_1 H_1 = M_1$, and M_1 and E_1 depend only on x . We show that $M_1 > 0$ and $E_1 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_1 and E_1 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_1 > 0$ and $E_1 > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . Hence $M_1 > 0$ and $E_1 > 0$, so (11) gives $4Q_1 \geq E_1 > 0$, and $Q_1 H_1 = M_1 > 0$ gives $H_1 > 0$.

The sign of H_2 . Since $F(x, y) = 0$, (12) gives $Q_2 H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -4$, write $x = -4 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $-3 \leq x \leq 0$. For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = -2$ and $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with such x . For $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{61}$ for every y , as one checks for the 61 residues of y modulo 61, so (1) has no solution with this x . Hence $M_2 > 0$ and $E_2 > 0$, so (13) gives $36Q_2 \geq E_2 > 0$, and $Q_2 H_2 = M_2 > 0$ gives $H_2 > 0$.

The sign of H_3 . Suppose that $H_3 \leq 0$. Since $H_3 = -y - 1$, this means $y \geq -1$. By (14), F is $x^4 + L_3 x + Q_3$, where L_3 and Q_3 depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_3 and Δ_3 as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_3 > 0$ and $\Delta_3 > 0$. For $y = -1$ and $y = 0$ we also have $Q_3 > 0$ and $\Delta_3 > 0$. Hence $Q_3 > 0$ and $\Delta_3 > 0$, so Lemma 6 with $a = 1$, $b = L_3$, $c = Q_3$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $H_3 > 0$.

The sign of H_4 . Suppose that $H_4 \leq 0$. Since $H_4 = -y - 1$, this means $y \geq -1$. By (16), F is $x^4 + L_4x + Q_4$, where L_4 and Q_4 depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_4 and Δ_4 as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_4 > 0$ and $\Delta_4 > 0$. For $y = -1$ and $y = 0$ we also have $Q_4 > 0$ and $\Delta_4 > 0$. Hence $Q_4 > 0$ and $\Delta_4 > 0$, so Lemma 6 with $a = 1$, $b = L_4$, $c = Q_4$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $H_4 > 0$.

By these signs, $A_1 < 0$, $B_1 < 0$, $A_2 > 0$, and $A_3 > 0$.

Hence Theorem 2(i) gives $(A_1, B_1)_\infty = -1$ (as $A_1 < 0$ and $B_1 < 0$), $(A_2, B_2)_\infty = 1$ (as $A_2 > 0$), and $(A_3, B_3)_\infty = 1$ (as $A_3 > 0$).

Step 2: primes outside S . Let $S = \{2, 3, 37\}$ be the set of primes dividing 2, the constant 1 in the $d_k = cH_{k'}$, $d_3 = 1$ and $d_4 = 2$, the right-hand side of (6), the right-hand side of (7), $V_3 = 1$, and the right-hand side of (8), and the right-hand side of (9). Let $p \notin S$ be an odd prime and $j \in \{1, 2, 3\}$; we show that $(A_j, B_j)_p = 1$. If $p \nmid A_j B_j$, this follows from Theorem 2(ii) with $\alpha = \beta = 0$. For $j = 1$: If $p \mid A_1$, then $p \mid H_1$, so $p \nmid H_2$ by (9) and $p \nmid B_1$; as p does not divide both U_1 and V_1 by (6), Lemma 3 with $(a, b, r, c, t) = (A_1, B_1, U_1, V_1, T_{A_1})$ and (18) gives $(A_1, B_1)_p = 1$; if $p \mid B_1$, then $p \mid H_2$, so $p \nmid H_1$ by (9) and $p \nmid A_1$; as p does not divide both U_2 and V_2 by (7), Lemma 3 with $(a, b, r, c, t) = (B_1, A_1, U_2, V_2, T_{B_1})$ and (18) gives $(B_1, A_1)_p = 1$, which equals $(A_1, B_1)_p$ by Theorem 2(v). For $j = 2$: Here $p \nmid B_2$ as $p \notin S$. If $p \mid A_2$, then, as $p \nmid V_3$, Lemma 3 with $(a, b, r, c, t) = (A_2, B_2, U_3, V_3, T_{A_2})$ and (19) gives $(A_2, B_2)_p = 1$. For $j = 3$: Here $p \nmid B_3$ as $p \notin S$. If $p \mid A_3$, then, as p does not divide both U_4 and V_4 by (8), Lemma 3 with $(a, b, r, c, t) = (A_3, B_3, U_4, V_4, T_{A_3})$ and (20) gives $(A_3, B_3)_p = 1$.

Step 3: the primes in S . By Step 1, the A_j and B_j are non-zero. If $T_{A_j} \neq 0$, then (18)–(20) and Lemma 4 give $(A_j, B_j)_v = (T_{A_j}, B_j)_v$, and if $T_{B_j} \neq 0$, they give $(A_j, B_j)_v = (B_j, A_j)_v = (T_{B_j}, A_j)_v = (A_j, T_{B_j})_v$, using Theorem 2(v). For a non-zero integer X and a prime p , we write $X \in p^{2i}(\pm)$ or $X \in p^{2i+1}(\pm)$, when p is odd, if $X = p^j u$ with $p \nmid u$, $(u/p) = \pm 1$, and j even or odd, respectively; and we write $X \in 2^{2i} \cdot u$ or $X \in 2^{2i+1} \cdot u$ if $X = 2^j u'$ with $u' \equiv u \pmod{8}$ and j even or odd. If $X \not\equiv 0 \pmod{p^e}$ and $v_p(X) \leq e - 1$, or $v_p(X) \leq e - 3$ if $p = 2$, then this class is determined by X modulo p^e . By Theorem 2(ii) and (iii), the classes of X and Y determine $(X, Y)_p$.

The prime 2. Checking the 16^2 residue pairs modulo 16, we find that (1) has exactly 16 solutions modulo 16, and that at each of them, for each j , one of the cases listed for $p = 2$ and j in Table 1 holds; moreover, the combination of these cases is one of the combinations listed there. Each case gives the stated value of $(A_j, B_j)_2$. Hence $\sigma_2 \in V_2$, with V_2 as in Step 4.

The prime 3. Checking the 3^2 residue pairs modulo 3, we find that (1) has exactly 2 solutions modulo 3, and that at each of them, for each j , one of the cases listed for $p = 3$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_3$. Hence $\sigma_3 \in V_3$, with V_3 as in Step 4.

The prime 37. Checking the 37^2 residue pairs modulo 37, we find that (1) has exactly 33 solutions modulo 37, and that at each of them, for each j , one of the cases listed for $p = 37$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{37}$. Hence $\sigma_{37} \in V_{37}$, with V_{37} as in Step 4.

Step 4: contradiction. For $p \in S$ and $p = \infty$, put

$$\sigma_p = ((A_1, B_1)_p, (A_2, B_2)_p, (A_3, B_3)_p).$$

By Theorem 2(iv), applied to each j , and by Step 2, the componentwise product of the σ_p with $p \in S$ is σ_∞ , which is $(-1, 1, 1)$ by Step 1. By Step 3, $\sigma_p \in V_p$, where

$$V_2 = \{(1, 1, 1), (1, 1, -1), \\ (1, -1, 1), (1, -1, -1)\},$$

$$\begin{aligned}
&(-1, -1, 1), (-1, -1, -1)\}, \\
V_3 &= \{(1, 1, 1)\}, \\
V_{37} &= \{(1, 1, 1)\}.
\end{aligned}$$

The componentwise products of one vector from each of V_3 and V_{37} are $(1, 1, 1)$, so σ_2 would have to be $(-1, 1, 1)$, but this vector is not in V_2 . This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The cases in Step 3. In each case, the symbol $(A_j, B_j)_p$ is computed as the stated symbol, whose entries lie in the stated classes; an entry “any” means that the other entry is a unit square, so that Lemma 5 applies.

	symbol	first entry	second entry	value
$p = 2$, solutions modulo 16				
$j = 1$				
1	(A_1, B_1)	$2^{2i} \cdot 3$	$2^{2i+1} \cdot 1$	-1
2	(A_1, B_1)	$2^{2i} \cdot 3$	$2^{2i+1} \cdot 3$	1
3	(A_1, B_1)	$2^{2i} \cdot 3$	$2^{2i+1} \cdot 5$	-1
4	(A_1, B_1)	$2^{2i} \cdot 3$	$2^{2i+1} \cdot 7$	1
5	(A_1, B_1)	$2^{2i+1} \cdot 1$	$2^{2i+1} \cdot 7$	1
6	(A_1, B_1)	$2^{2i+1} \cdot 3$	$2^{2i+1} \cdot 5$	1
7	(A_1, B_1)	$2^{2i+1} \cdot 5$	$2^{2i+1} \cdot 3$	1
8	(A_1, B_1)	$2^{2i+1} \cdot 7$	$2^{2i+1} \cdot 1$	1
9	(T_{A_1}, B_1)	$2^{2i} \cdot 1$	any	1
$j = 2$				
1	(A_2, B_2)	$2^{2i} \cdot 1$	$2^{2i} \cdot 7$	1
2	(A_2, B_2)	$2^{2i} \cdot 3$	$2^{2i} \cdot 7$	-1
3	(A_2, B_2)	$2^{2i} \cdot 5$	$2^{2i} \cdot 7$	1
4	(A_2, B_2)	$2^{2i} \cdot 7$	$2^{2i} \cdot 7$	-1
$j = 3$				
1	(A_3, B_3)	$2^{2i} \cdot 1$	$2^{2i+1} \cdot 7$	1
2	(A_3, B_3)	$2^{2i} \cdot 3$	$2^{2i+1} \cdot 7$	1
3	(A_3, B_3)	$2^{2i} \cdot 5$	$2^{2i+1} \cdot 7$	-1
4	(A_3, B_3)	$2^{2i} \cdot 7$	$2^{2i+1} \cdot 7$	-1
combinations of cases ($j = 1, j = 2, j = 3$) and the vector σ_2				
$(1, 2, 2) \mapsto (-1, -1, 1)$		$(2, 1, 1) \mapsto (1, 1, 1)$		
$(3, 4, 4) \mapsto (-1, -1, -1)$		$(4, 3, 3) \mapsto (1, 1, -1)$		
$(5, 4, 4) \mapsto (1, -1, -1)$		$(6, 2, 2) \mapsto (1, -1, 1)$		
$(7, 4, 4) \mapsto (1, -1, -1)$		$(8, 2, 2) \mapsto (1, -1, 1)$		
$(9, 1, 1) \mapsto (1, 1, 1)$		$(9, 3, 3) \mapsto (1, 1, -1)$		
$p = 3$, solutions modulo 3				
$j = 1$				
1	(A_1, B_1)	$3^{2i}(+)$	any	1
2	(T_{A_1}, B_1)	$3^{2i}(+)$	$3^{2i}(+)$	1
$j = 2$				
1	$3 \nmid A_2 B_2$			1
$j = 3$				
1	$3 \nmid A_3 B_3$			1

	symbol	first entry	second entry	value
$p = 37$, solutions modulo 37				
	$j = 1$			
1	$37 \nmid A_1 B_1$			1
2	(A_1, T_{B_1})	$37^{2i}(-)$	$37^{2i}(-)$	1
3	(A_1, T_{B_1})	$37^{2i}(+)$	$37^{2i}(+)$	1
4	(T_{A_1}, B_1)	$37^{2i}(-)$	$37^{2i}(+)$	1
	$j = 2$			
1	$37 \nmid A_2 B_2$			1
	$j = 3$			
1	$37 \nmid A_3 B_3$			1